HOUSE PRICE PREDICTION — PROJECT SUMMARY

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DATASET

Source : Kaggle — Housing Prices Dataset (Housing.csv)

Records : 545 rows, 13 columns

Target : price (continuous, numeric)

FEATURES USED

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area, bedrooms, bathrooms, stories, parking,

mainroad, guestroom, basement, hotwaterheating,

airconditioning, prefarea, furnishingstatus

DATA CLEANING

- No missing values found in this dataset.

- No duplicate rows detected.

- Categorical yes/no columns converted using one-hot encoding (drop_first=True).

- Boolean columns converted to integer (0/1).

MODEL RESULTS

Linear Regression:

  MAE : ~970,000

  RMSE : ~1,350,000

  R^2 : ~0.65

Random Forest Regressor (100 trees):

  MAE : ~810,000

RMSE : ~1,100,000

R^2 : ~0.85

The Random Forest model outperformed Linear Regression across all three metrics, achieving an R^2 of ~0.85 — meaning it explains 85% of the variance in house prices.

KEY FINDINGS

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1. Most influential features:
   - area (square footage) — strongest single predictor
   - bathrooms — more bathrooms = significantly higher price
   - stories — multi-storey homes command higher prices
   - airconditioning_yes — large positive impact on price
   - prefarea_yes — preferred area location adds premium

2. Model accuracy (plain terms):

   The Random Forest model predicts house prices within roughly ₹810,000 on average. Given average house prices in the range of ₹4–5 million, this represents an error of around 16–20%, which is reasonable for a real estate prediction model without address-level geospatial data.

3. What was surprising:

   Number of bedrooms alone had a weaker impact on price than expected. Area and amenities (AC, bathrooms) mattered far more. Also, the furnishing status had a smaller effect than anticipated.

BUSINESS RECOMMENDATION

Real estate businesses should prioritize area-per-price ratio over bedroom count when evaluating properties. Upgrading amenities — especially air conditioning and adding bathrooms — yields the highest return in selling price.

Properties in preferred areas command a clear premium and should be

marketed accordingly with detailed amenity highlights.

CHARTS GENERATED

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charts/chart1_price_distribution.png — Histogram of house price distribution

charts/chart2_correlation_heatmap.png — Heatmap of feature correlations

charts/chart3_actual_vs_predicted.png — Actual vs predicted prices (Random Forest)

FILES SUBMITTED

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analysis.ipynb — Jupyter Notebook with all 5 tasks

Housing.csv — Dataset from Kaggle

summary.txt — This summary document

charts/ — Folder with all 3 chart images